Accurate Distances, Imperfect Geometry: Repeated Language-Model Judgments across 100 Capitals

Abstract

Accurate distance estimates need not describe a consistent geography. We collected 148,500 completed responses from Gemini 3.5 Flash for 4,950 pairs of 100 national capitals, using ten requests per pair under English, Arabic and written Chinese instructions. Pair-median mean absolute errors are 216–234 km and rank correlations exceed 0.998, yet 10.7–13.3% of triangles violate the triangle inequality. Controls preserving every pair's absolute error but randomizing its sign produce 11.5–12.3% violations on average: the violation rate alone does not establish unusual spatial incoherence. Spherical fits place capitals 150–158 km from reference geography. Between-condition fitted shifts are 79–88 km; conditional randomization concerns response distributions, not population median maps or isolated language effects. The benchmark connects accuracy, coherence and repeated-response variation. Generalization beyond these fixed prompts, entities and collection period remains untested.

Keywords: geographic representation; language models; spatial consistency; spherical reconstruction; uncertainty; multilingual evaluation.

Running title: Geography from model judgments

1 Introduction

What geographic configuration best reconciles a language model's answers about distances between places? Answering this question requires separating three properties: agreement with real distances, consistency of the answers as a metric, and variation across repeated responses. A model can recall many distances accurately while making mutually incompatible judgments; it can also repeat an incorrect answer with little variation.

Geographic reconstruction from language has substantial precedent. Louwerse and Zwaan recovered geographic structure from distributional text statistics [12]. Bhandari et al. elicited numerical distances among 93 contiguous-US cities, reconstructed coordinates with two-dimensional multidimensional scaling (MDS), and evaluated aligned positions against geographic controls [2]. Their use of five-beam decoding differs from the repeated stochastic sampling studied here. Karimi and Janowicz supplied illustrative English–Persian distance comparisons using Google Maps routing distances as their reference [9]. Those routing examples motivate multilingual investigation but use a different target from the geodesics in this study.

More recent work evaluates geographic distortions through tokenization, country knowledge and semantic distances [5], while MultiGlobeQA covers 14 spatial-function families across 17 languages [3]. These studies establish that geographic evaluation and multilingual coverage are active research areas. Appendix A compares their measurement targets with this study.

The present contribution is a global repeated-judgment benchmark that connects response variation, metric inconsistency, spherical reconstruction, and conditional comparisons between translated prompts. A spherical fit accommodates global arc distances; retaining ten observations per pair exposes differences hidden by a single aggregate. An error-sign control additionally asks how much metric inconsistency is compatible with the observed absolute errors. These are empirical extensions of established reconstruction methods, not a new embedding algorithm or a claim to reveal a hidden neural representation. The core question is empirical: how do accurate pairwise judgments and imperfect global consistency coexist, and how much does the resulting configuration vary between these three prompt conditions?

Here, a cognitive map is an operational description of observable spatial judgments. Activation-based work investigates internal representations directly [6]; behavioral planning evaluations such as CogEval ask whether relational knowledge supports navigation and planning [13]. A geometry fitted to API outputs establishes neither of those mechanisms. Memorized facts, heuristics, and internal computation remain compatible explanations.

The numerical results use the 100-capital English, Arabic, and Chinese collection. All analyses are exploratory, with no preregistration or independent replication across phrasings or dates.

2 Data and elicitation

2.1 Entities and geographic reference

The purposive sample contains 100 national capitals across the six inhabited continents. It is neither a random sample of countries nor the complete set of national capitals. Appendix G lists the complete sample. Natural Earth populated-place coordinates provide the reference city points [14, 15]; the dataset supplement preserves source URLs, content hashes, overrides, and selection decisions. Examples include Pretoria as South Africa's executive capital, Sucre as Bolivia's constitutional capital, Bern as Switzerland's federal city, and Yamoussoukro as Côte d'Ivoire's political capital. One reference point per capital cannot capture all choices of city center or administrative extent.

For each of $P = 100 \times 99/2 = 4{,}950$ unordered pairs, reference distance t_p is the WGS84 inverse geodesic computed with GeographicLib [10]. These are surface distances on an ellipsoid. Map projections are used only for display and planar alignment, never as the geographic distance benchmark.

2.2 Three matched prompt conditions

The requested and returned model identifier is `gemini-3.5-flash`. All conditions use temperature 1.0, top-p 0.95, a 128-token output ceiling, minimal thinking, no specified seed, and an empty system prompt. Minimal thinking is an API setting, not evidence of zero internal computation. Every pair receives ten separate single-turn requests. The three conditions change the complete instructions while retaining canonical English city and country names and the same pair ordering.

The English prompt asks for a great-circle distance and one numerical answer, followed by a numeric-format instruction; Appendix C gives its text. Arabic uses written standard Arabic. The Chinese condition uses written standard Chinese in simplified characters; the website's historical label "Mandarin Chinese" does not imply a speech experiment. The translations were assistant-authored and have not undergone independent native-speaker validation. They are naturally worded rather than literal substitutes: the Arabic and Chinese versions explicitly explain shortest surface distance. Consequently, language, phrasing, and explanatory wording are not separately identified.

Collection occurred on 8 September 2026. The runner dispatched ten requests for a pair before advancing through canonical pair order; conditions ran concurrently with rate limiting. Median within-pair first-attempt dispatch spans were approximately 0.87–0.88 seconds. These are separate requests, not verified independent draws from a stationary provider. Temporal clustering, caching, routing, and unreported model changes remain possible sources of dependence. The response identifier supplied by Google is a model alias, not a frozen checkpoint guarantee.

2.3 Response validation and retained evidence

The collection used unconstrained text generation, *not provider-enforced output schemas*. The recorded parser accepts a whole response consisting of ASCII digits with an optional period-delimited fractional part. It does not extract a convenient number from prose, interpret decimal commas, or repair malformed text. Nonpositive, nonfinite, and greaterthan-20,040 km values are flagged. This upper bound is a documented plausibility rule, not a value supplied to the model. New schema-constrained conditions introduced later in the software are outside this study.

The 148,500 completed sample slots yielded 148,491 valid values. There were 148,504 attempts: four additional transport retries in Arabic, all followed by a terminal response. Invalid content was terminal rather than retried until valid. Five numerical estimates above the range threshold occurred for Buenos Aires–Hanoi: four in English and one in Arabic. Four Chinese responses for Brussels–Paris included a trailing backtick or underscore. All raw answers, classifications, retries, and provider payloads remain available. Replaying the recorded parser on saved responses produced zero mismatches.

Descriptive matrices use every pair's available valid observations. Tests of pair-median disagreement and accuracy use the same 4,948 complete pairs in all three conditions, excluding Buenos Aires–Hanoi and Brussels–Paris. This removes 51 otherwise valid observations in addition to the nine invalid values. Map randomization uses all 4,950 pairs and preserves each condition's valid-response count. These denominators are kept distinct throughout.

3 Analysis

3.1 Accuracy, variability, and consistency

For pair p, condition ℓ, and replicate r, let $x_{p\ell r}$ be an accepted distance and $m_{p\ell}$ its sample median. The principal global accuracy statistic is *pair-median MAE*,

$$A_\ell = P^{-1} \sum_p |m_{p\ell} - t_p|. \tag{1}$$

RMSE, rank correlations, signed error, and relative error provide complementary summaries. Mean, median, 10%-trimmed mean, and Huber aggregation remain available in the exports; the manuscript consistently uses medians for its primary reconstructions. Within-pair standard deviations, coefficients of variation, and repeated-value counts describe sampling stability without assuming Gaussian responses.

The symmetric matrix D_ℓ has zero diagonal and the pair medians off diagonal. Symmetry is imposed by the unordered design, not tested by opposite-direction queries. All $\binom{100}{3} =$ 161,700 triples are examined. A triple violates the triangle inequality when its longest side exceeds the sum of the other two by more than 10^{-7} km. We report the percentage of violated triples and their excess magnitudes. Three-nearest-neighbor preservation is measured both in the input distance matrix and after reconstruction, relative to the WGS84 matrix.

3.2 An absolute-error-preserving control

A high triangle-violation rate can arise from otherwise accurate but imperfect pair estimates. To separate the magnitude of estimation error from its signed arrangement, let $a_{p\ell} = |m_{p\ell} - t_p|$ and construct artificial distances $t_p + s_{p\ell} a_{p\ell}$. Signs are independently equiprobable where both choices lie in $(0, 20{,}040]$ km; otherwise the sole admissible sign is retained. Thus each pair's absolute error, and consequently MAE, RMSE and absolute relative error, are preserved. The control changes signs, not the identity or magnitude of the pair errors.

For each condition, 999 seeded assignments are evaluated on all triples. Their central 95% ranges describe the artificial assignments, not confidence intervals or a validated sampling law for LLM responses. The control assumes independent error signs except for the range constraint; it preserves neither systematic signed bias nor cross-pair dependence. It is a conditional descriptive diagnostic selected after the main analysis, not a test identifying a cognitive mechanism. Artificial distances are stored separately from observations, and no distributional form is assumed for the ten model responses.

3.3 Recovering and aligning geometry

Classical MDS retains leading nonnegative eigencomponents of $-\frac{1}{2} J D^{\circ 2} J$, where $J = I - \mathbf{1}\mathbf{1}^{\mathsf{T}}/N$. Metric and non-metric MDS use scikit-learn's four-start SMACOF implementation, seed 42, up to 1,000 iterations, and tolerance 10^{-7} [11, 17]. The non-metric fit optimizes monotone disparities and has a distinct stress definition. We also fit metric

Euclidean embeddings in dimensions 1 through 10 to both judgments and reference geodesics. Comparisons on kilometer distances use a common target-normalized residual,

$$S(D, \widehat{D}) = \sqrt{\frac{\sum_{i<j}(\widehat{D}_{ij} - D_{ij})^2}{\sum_{i<j} D_{ij}^2}}. \tag{2}$$

Non-metric disparity Stress-1 is reported separately, not ranked against this residual.

The spherical fit places unit vectors u_i on a sphere of fixed radius $R = 6371.0088$ km and minimizes

$$\sum_{i<j}\{R \arccos(u_i^{\mathsf{T}} u_j) - D_{ij}\}^2. \tag{3}$$

Initialization comes from the three leading components of $\cos(D/R)$, with subsequent starts perturbed by seeded Gaussian angular noise of standard deviation 0.3 radians. Sparse analytic derivatives support least-squares optimization with at most 1,000 function evaluations and 10^{-9} convergence tolerances. The published fits use four starts. A twelve-start diagnostic per judgment matrix and a four-start WGS84 control record convergence, objectives and aligned-position differences for every start.

A single orthogonal Procrustes transform aligns fitted unit vectors with reference geography [16]. It permits global rotation or reflection at fixed radius. Reference coordinates enter after fitting; individual capitals are not pulled toward known positions. Planar display alignment additionally permits translation and one global scale. Mean aligned capital displacement is descriptive and includes reference-point uncertainty and mismatch between the fitted sphere and the WGS84 ellipsoid.

An exact spherical distance matrix in the valid arc domain must yield a positive-semidefinite cosine Gram matrix of rank at most three. The supplement therefore records the complete spectrum, negative eigenmass, positive mass beyond rank three, numerical tolerance, and distance-domain checks. Negative eigenmass alone is insufficient. Ellipsoidal reference geodesics need not satisfy exact spherical compatibility, so they are analyzed as a control rather than assigned zero spherical residual.

Coastlines are visualized by interpolating the fitted capital displacements through a Delaunay mesh in an equirectangular display, with frame controls and longitude-seam handling. Piecewise-affine interpolation is continuous within the mesh, but can fold triangles and alter topology. Coastline shape is not directly elicited or tested. The fitted capitals and their distance relationships are the measured geometric objects.

3.4 Conditional comparisons between prompt conditions

For each of the three condition pairs, mean absolute median disagreement is

$$T_{ab} = P^{-1} \sum_{p} |m_{pa} - m_{pb}|. \tag{4}$$

We compare T_{ab} and the absolute pair-median MAE difference with 4,999 random assignments obtained by pooling the twenty responses within each complete pair and splitting them into two groups of ten. Replicate numbers are not matched across conditions. The strong null is equality of response distributions within every pair, under independent requests. Upper-tail disagreement and absolute-value accuracy statistics use $(b + 1)/(B + 1)$ Monte Carlo probabilities [19]; Holm adjustment covers these six tests [7].

A separate exploratory map statistic is mean geodesic displacement between two independently fitted spherical configurations after global orthogonal alignment. For 999 within-pair label permutations per comparison, we rebuild both median matrices and refit both maps with the same four starts. Each permutation preserves the observed valid sample count in each condition, including incomplete pairs. All observed fits reproduce the published results within one metre and every randomized fit converges. Holm adjustment covers the three map comparisons. Saved null statistics permit direct verification of these probabilities.

Both procedures test a common *response-distribution* baseline. Different dispersions can cause rejection even when population median distances are identical. Map-displacement randomization inherits this limitation: its geometric statistic does not turn the null into equality of population median maps. Neither procedure isolates language from translation wording. The separate six-test and three-test corrections do not provide study-wide control over earlier exploratory choices.

3.5 Exploratory regional comparisons

The global analysis evaluates a median of ten responses. The regional analysis instead evaluates *mean single-response absolute error.* Two transparent, post hoc groups are used: the 15 sampled capitals in a frozen 22-member Arab States list [8], and the two named cases Beijing and Singapore. The latter reflects institutional standard-language association for China and documented Mandarin use in Singapore [21, 20]; it is not an exhaustive classifier of Chinese-speaking places. Appendix D lists the groups and their sources. These two definitions represent different regional and linguistic constructs, so their effect sizes are not a common scale of cultural affinity.

A pair is associated with a group if *at least one* endpoint belongs to it. With $e_{p\ell r} = |x_{p\ell r} - t_p|$, define pairwise gain $g_{p\ell} = \bar{e}_{p,\text{en}} - \bar{e}_{p\ell}$. Associated-pair gain G_H averages g on associated pairs; G_O averages it elsewhere; the interaction is $I = G_H - G_O$. Positive gain means lower error under the translated prompt. A positive interaction can mean a smaller disadvantage, rather than an absolute benefit.

We report available-pair and capital-omission sensitivities and balance both groups descriptively to pooled WGS84 distance quintiles. This standardization addresses distance composition only, not unmeasured regional familiarity or a causal effect. Regional conclusions use the observed gains and sensitivities, without hypothesis-test or confidence-interval claims. An empirical-bootstrap audit using 9,999 draws is retained in Appendix E for methodological transparency. Its rare-tail calibration scenario produced only 3.3–7.5% associated-gain coverage and 100% familywise rejection. Matching the sample variance did not repair unseen tails. The actual response law is unknown, so these diagnostics neither prove that it has rare tails nor justify treating the resulting probabilities as reliable evidence of a regional advantage.

4 Results

4.1 High rank accuracy coexists with metric inconsistency

The three conditions track global geography closely (Figure 1; Table 1). English has the lowest observed pair-median MAE, 215.9 km; Arabic has 234.0 km and Chinese 222.0 km. All Spearman correlations exceed 0.998. These are condition-specific observations from one model configuration, not an inferential ranking of model families or a guarantee of future performance.

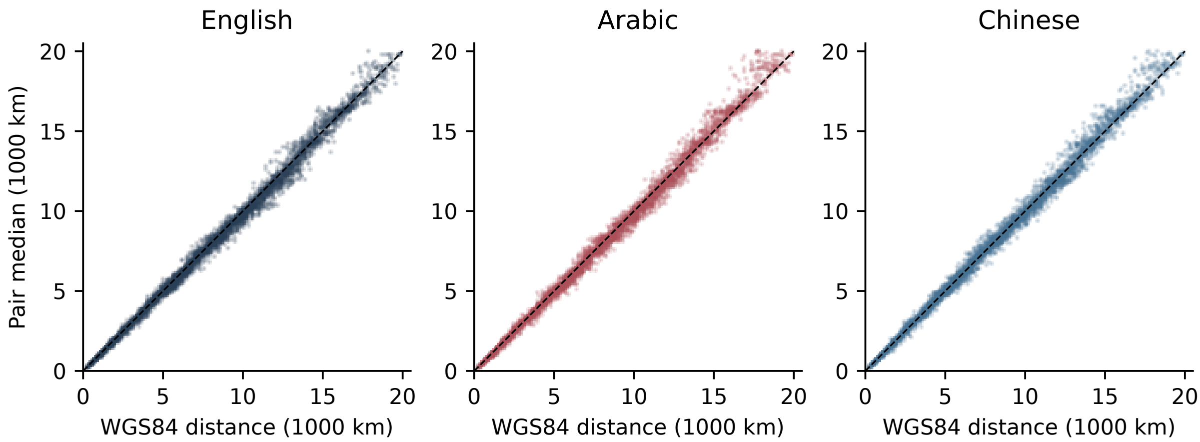

Figure 1: All 4,950 pair medians per condition against WGS84 geodesics. The dashed diagonal is perfect accuracy. A wide global distance range makes the association visually strong even when errors reach hundreds of kilometers.

Table 1: All-pair descriptive accuracy. Each condition has 49,500 completed sample slots. MAE and RMSE concern pair medians; triangles is the percentage of all 161,700 triples with a violation.

Instructions	Valid values	MAE (km)	RMSE (km)	Spearman	Triangles (%)
English	49,496	215.9	341.3	0.99825	10.69
Arabic	49,499	234.0	372.2	0.99809	13.30
Chinese	49,496	222.0	354.9	0.99810	10.74

High rank agreement does not imply a metric: 10.69% of English, 13.30% of Arabic, and 10.74% of Chinese triples violate the triangle inequality. Among violated triples, the mean excess is 322.8, 421.3, and 350.4 km, respectively. The corresponding WGS84 reference has no violations. These counts describe the input judgments; the fitted spherical distances obey metric constraints by construction.

The error-sign controls give 11.52%, 12.30% and 11.85% mean violation rates for English, Arabic and Chinese (Table 2). English and Chinese are below their respective central control ranges, whereas Arabic is above its range. The observed arrangement of signed errors therefore differs from these independent-sign controls in direction as well as magnitude; a violation rate near 10% is not, by itself, evidence of exceptionally incoherent geographic knowledge. Neither this pattern nor its relation to prompt condition identifies the source of the errors.

Table 2: Conditional error-sign controls with identical pairwise absolute errors. Rates and central control ranges are percentages of all triples. Ranges summarize 999 artificial assignments, not sampling confidence intervals. Only 3, 2 and 2 pairs, respectively, require a fixed sign to remain in the reporting domain.

Instructions	Observed	Control mean	Central control range
English	10.69	11.52	11.15–11.92
Arabic	13.30	12.30	11.91–12.71
Chinese	10.74	11.85	11.46–12.23

Repeated outputs contain both ties and variation (Table 7). Among 4,949 pairs with ten valid responses in each individual condition, 139 English, 123 Arabic, and 64 Chinese pairs give ten identical values. Mean within-pair standard deviations are 116.4, 125.2, and 121.5 km. This discreteness argues against modeling the recorded outputs as unrounded continuous Gaussian draws, but does not exclude rounded latent distributions or identify their tails. Ten observations cannot establish modality or equate low response variance with epistemic certainty.

4.2 A recognizable sphere with measurable residuals

The reconstructions retain recognizable global geography (Figure 2), but their residual stress is much larger than the ellipsoidal reference's spherical fitting floor (Table 3). The twelve-start diagnostic fits place capitals about 150–158 km from their reference positions on average, versus 9.5 km for the WGS84 control. The reference displacement is not subtracted from model displacements: the vectors vary by capital and do not define a scalar correction.

Input judgments preserve roughly 96% of the reference three-nearest-neighbor sets; fitted spherical positions preserve about 92–93%. Enforcing global geometry therefore trades some local neighborhood fidelity for a coherent metric. Fit and neighborhood statistics are in-sample. They do not establish unique identification, planning ability, or performance on unqueried cities.

Table 3: Spherical geometry and reference control. Diagnostic fits use twelve starts per judgment matrix and four for WGS84. Stress follows Eq. 2. Shift is mean displacement from actual capitals. NN columns give input and fitted three-neighbor preservation. Excess averages violated input triples only; a dash denotes no violated triples.

Input	Stress	Shift (km)	Input NN (%)	Fitted NN (%)	Excess (km)
WGS84 control	0.00043	9.5	100.0	100.0	—
English	0.02992	157.9	96.3	92.3	322.8
Arabic	0.03503	150.1	96.7	92.3	421.3
Chinese	0.03206	154.2	96.0	92.7	350.4

All 40 diagnostic starts converged. Increasing from four to twelve starts changes the aligned judgment maps by only approximately 1–3 metres per capital on average. This supports stability for the tested starts, not a proof of the global optimum. Full spectra show both negative cosine eigenmass and positive mass beyond rank three in the judgments. The WGS84 control has a small spectral residual too, as expected from the ellipsoid–sphere mismatch.

The Euclidean stress curves follow a broadly similar decline for judgments and WGS84 distances (Figure 3). Even correct global arc distances retain Euclidean residuals: three-dimensional chord distances are not surface arcs. Consequently, residual stress at higher dimensions is not direct evidence of additional semantic dimensions. No intrinsic dimensionality is selected. Appendix F.1 distinguishes common distance residuals from non-metric disparity stress.

4.3 Small map shifts relative to a conditional baseline

On the common 4,948 complete pairs, English–Arabic median disagreement is 141.3 km and English–Chinese disagreement is 157.7 km. Arabic–Chinese disagreement is 165.2 km (Table 8). Complete-case pair-median MAEs are 215.8, 233.8, and 222.0 km for English,

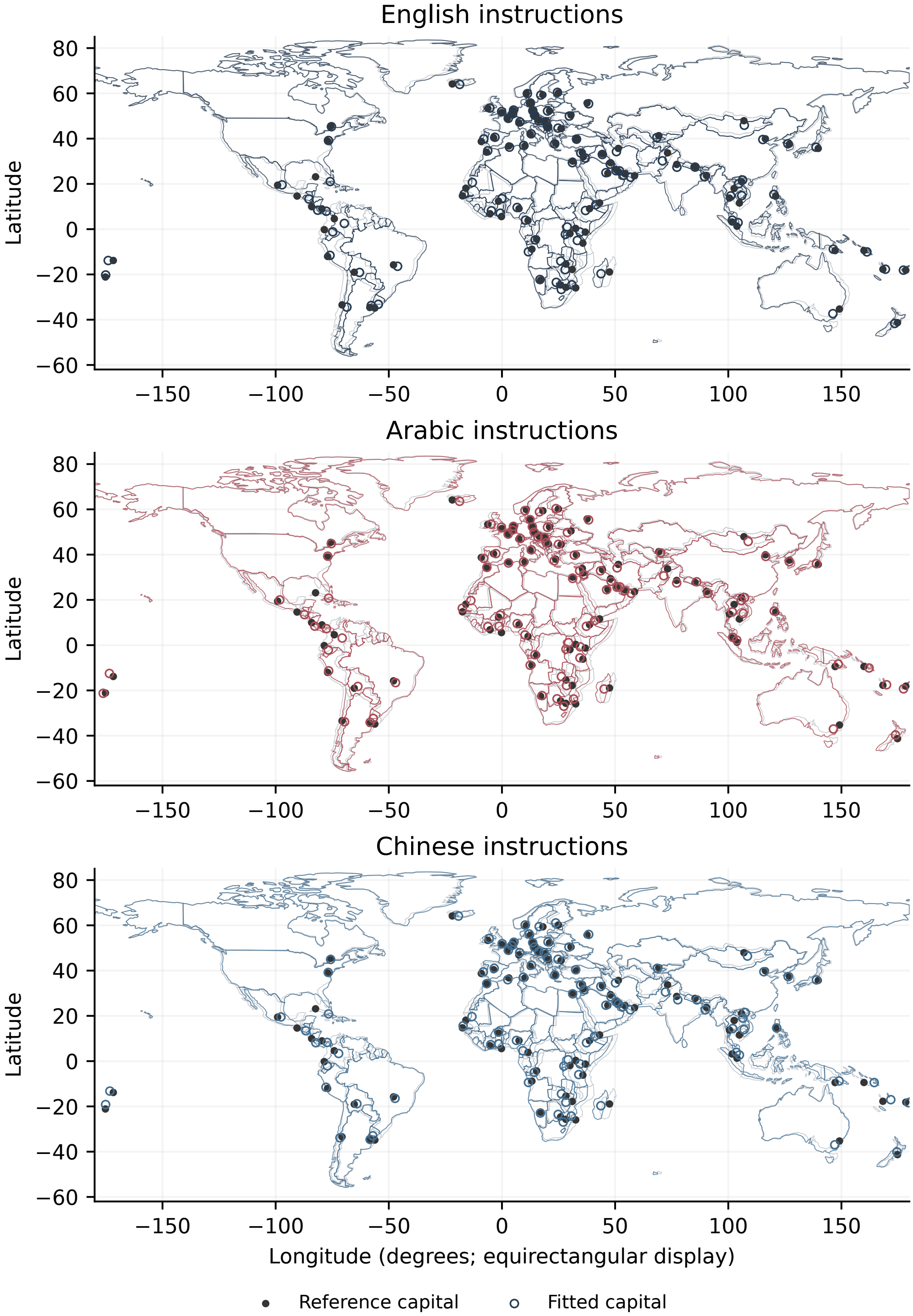

Figure 2: The three published four-start, median-based spherical reconstructions. Gray outlines and filled points show reference geography; colored coastlines and open points show the deformation and fitted capitals. Displacement is unamplified. All panels use the same equirectangular display; its projection distortion is distinct from the fitted capital shifts. Coastline interpolation is illustrative.

Arabic, and Chinese. Thus Chinese and English differ by 157.7 km in their individual pair medians while their aggregate MAEs differ by only 6.3 km.

Under the assumed within-pair independence and exchangeability model, all six distance statistics exceed the shuffled baseline, with adjusted Monte Carlo $p = 0.0012$. No randomized statistic exceeded its corresponding observed magnitude among 4,999 draws. The repeated adjusted value reflects simulation resolution and multiplicity correction, not equal effect sizes or an exact underlying probability.

The observed mean aligned capital shifts are 81.0 km for English–Arabic, 88.4 km for English–Chinese, and 78.5 km for Arabic–Chinese (Table 4). Their shuffled-label means range from 23.9 to 27.0 km. All three adjusted probabilities are 0.003, with zero exceedances among 999 permutations. Shifts of this size are subtle at world scale, explaining why the maps can look similar. Under the assumed within-pair exchangeability model, the observed shifts exceed the shuffled baseline; changes in dispersion can contribute. The collection design does not establish those exchangeability assumptions. They do not establish distinct population median maps.

Table 4: Map-displacement randomization using all pairs and their observed valid sample counts. The interval describes the central 95% of shuffled statistics; it is not a confidence interval for the observed map shift.

Conditions	Observed (km)	Null mean (km)	Null interval (km)	Adjusted p
English / Arabic	81.0	25.3	22.1–28.8	0.003
English / Chinese	88.4	23.9	21.3–26.8	0.003
Arabic / Chinese	78.5	27.0	23.8–30.2	0.003

4.4 Regional estimates do not show an absolute local advantage

The Arab States group has 1,380 associated complete pairs, 1,275 with only one associated endpoint and 105 with both. Mean single-response absolute error rises from 196.2 km under English to 205.1 km under Arabic, a gain of −8.9 km. On other pairs the Arabic disadvantage is larger, so the interaction is +13.1 km. This positive interaction describes a smaller disadvantage, not an accuracy benefit.

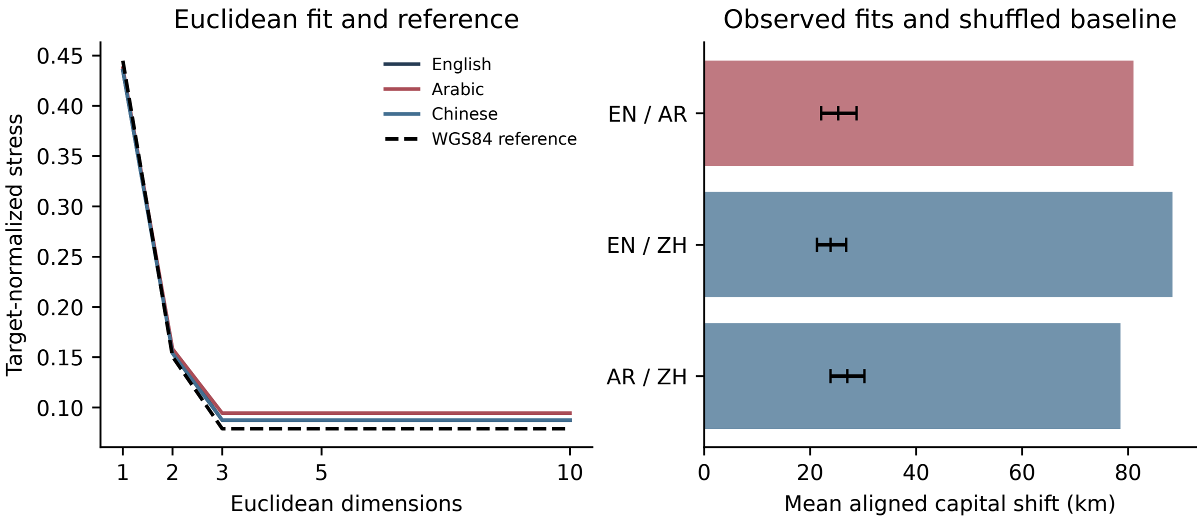

Figure 3: Left: metric Euclidean MDS residuals alongside the WGS84 baseline. Right: observed map shifts (bars) and shuffled-label means with central 95% null intervals (black markers). Solver variation can produce slight nonmonotonicity in the dimensionality curve.

The Beijing–Singapore group has 197 pairs: 196 with one associated endpoint and only one linking the two cases. Mean single-response error rises from 169.6 to 180.2 km under Chinese, a gain of -10.7 km. The interaction is -4.5 km. These group-specific observations do not support a local accuracy advantage in this collection. They also do not rule out advantages in other places or prompt formulations.

Table 5: Exploratory regional contrasts on complete pairs. Error columns average individual responses on associated pairs, unlike the pair-median MAE in Table 1. Gains and interactions are in km. These are descriptive estimates. The empirical-bootstrap audit and its calibration failure are reported separately in Appendix E.

Prompt	Pairs	English error	Local error	G_H	I
Arabic	1380	196.2	205.1	-8.9	+13.1
Chinese	197	169.6	180.2	-10.7	-4.5

Distance composition differs markedly: median reference distances are 5,451 km on Arab States pairs versus 8,474 km elsewhere, and 8,659 km on Beijing–Singapore pairs versus 7,562 km elsewhere. Standardizing both groups to the pooled distance-quintile distribution gives gains of -8.6 and -12.0 km and interactions of $+13.6$ and -6.0 km, respectively. This limited adjustment preserves the absence of an observed absolute gain. Capital omissions leave the Arab States gain between -12.0 and -6.6 km. The Chinese gain is -17.5 km when

Beijing is omitted and −3.9 km when Singapore is omitted, demonstrating heterogeneity concealed by the two-case average.

4.5 Sensitivity to exclusions

Retaining the five readable over-range numbers changes all-pair median MAEs to 216.12 km for English, 233.99 km for Arabic, and 222.02 km for Chinese (Appendix F). The ordering is unchanged. The malformed Chinese answers are not repaired or assigned values. Using all available pairs changes the regional interactions to +13.07 and −4.67 km while leaving associated-pair gains unchanged. Refitting after range retention shifts the English map by 0.90 km per capital on average (38.0 km maximum) and Arabic by 0.055 km (2.2 km maximum); Chinese is unchanged. The three between-condition map shifts change by at most 0.16 km on average. These checks quantify the recorded exclusions; they do not address unobserved tails, and no new significance claims are made from them.

5 Discussion

The main result is the coexistence of high geographic rank accuracy with imperfect metric consistency, interpreted against controls that preserve the observed errors. Similar violation rates arise after randomizing error signs, so much of the raw inconsistency is compatible with estimation errors of the measured size. The conditional differences from that control describe error arrangement, not an identified neural or cultural mechanism. Spherical reconstruction summarizes that inconsistency in a coherent configuration, making global structure visible while revealing a tradeoff in local neighborhood preservation. Reference controls matter: part of any spherical residual comes from fitting ellipsoidal surface distances, and Euclidean dimensionality curves also carry an Earth baseline.

The three prompt conditions produce different observed answers and fitted configurations. Their statistics exceed a shared response-distribution baseline conditional on the assumed within-pair exchangeability model. The effect sizes explain the apparent visual similarity: average between-condition capital shifts are roughly half the approximately 150-km displacements from reference geography and small relative to a world map. Statistical detectability and visual prominence answer different questions. A common-distribution rejection does not distinguish changes in central judgments from changes in sampling variability.

The regional analysis reinforces the distinction between an interaction and a benefit. Arabic has a smaller disadvantage on the chosen Arab States pairs than elsewhere, but still a larger observed error than English there. Chinese also shows larger error in its narrow two-case group. These findings are descriptive of the chosen operational sets. Regional geography, distance, prominence, multilingualism, and unknown training exposure remain intertwined. Coarse distance balancing does not identify a cultural or linguistic mechanism.

Several limitations delimit the evidence. The capitals were selected purposively, and the grouped analyses were formulated after earlier results were known. Each written language has one unvalidated translation with English entity names, one model alias, and one collection period. Closely clustered requests may share provider-side disturbances. Ten observations per pair can omit rare modes. The bootstrap calibration demonstrates that this last problem can defeat nominal interval coverage; more precise Monte Carlo probabilities do not repair missing tail information. The analysis does not estimate uncertainty over new countries, translations, dates, or model versions.

Future confirmatory work should prespecify the central estimands and comparison families, randomize languages and pairs within independently repeated time blocks, vary reviewed translations, and collect more repeats on a stratified subset. Localized entity names should form a separate factor. Stronger claims about population median maps need procedures justified for unequal response dispersions; behavioral navigation tasks would be needed to test use of the inferred geometry.

6 Data availability and research provenance

The public source repository, `https://github.com/timini/cognitive-atlas`, and public atlas, `https://timini.github.io/cognitive-atlas/`, provide responses, exact prompts, manifests, matrices, fitted coordinates and analysis code. The versioned dataset and computational release are identified by the experiment table in Appendix C and their checksums [1]. A full reproduction package and smaller supplement packages are available with the paper. A permanent archival DOI and project-wide reuse licenses have not yet been assigned; public downloadability does not itself grant reuse rights.

Builds verify fixed input and release hashes before generating tables and figures. Saved raw responses support parser replay, and numerical reproduction is checked with stated tolerances. Collection records include prompts, settings, dates and provider payloads, but do not capture dirty-source state or a collector snapshot for each resumed session. Historical analysis identities also omitted coastline and analysis-runtime inputs. These missing records prevent complete reconstruction of the historical execution environment; later checksums do not repair that limitation. Detailed computational provenance accompanies the release. Rebuilding the paper makes no model calls.

AI assistance

OpenAI Codex assisted with study implementation, collection orchestration, statistical and geospatial code, analyses, figures, translations, manuscript drafting and revision, and source checking. Ten academic-review agents and five publication-readiness agents supplied further feedback. These activities involved substantive generated material and interpretation, not only copyediting. Historical assistant model/version identifiers were not consistently recorded. Gemini 3.5 Flash was the measured system and is documented separately in the elicitation protocol.

Automated checks include parser replay, input verification, numerical invariants, control simulations, software tests and document rendering. They do not constitute independent human scientific assessment, native-speaker validation or proof of historical provider behavior. AI tools are not authors. Human author identities, contributions, funding and competing-interest declarations have not yet been supplied for submission.

Study materials

The experiment used model API responses and public geographic reference points; it collected no human-participant responses, private individual records or animal observations. This description does not substitute for any declaration required by a selected journal.

7 Conclusion

Across 100 capitals, Gemini 3.5 Flash's repeated distance judgments preserve global geographic order while violating exact metric consistency. Their fitted spherical maps are recognizable, quantitatively displaced from Earth, and sensitive to the observed prompt condition. Conditional randomization concerns a shared response-distribution baseline, not independently established population median maps. Error-sign controls show why the raw violation rate should not be interpreted as a distinctive cognitive defect. The data show no observed absolute regional advantage for the two translated prompts under the stated groups. The benchmark provides a reproducible account of elicited geography whose broader interpretation awaits replicated prompts, sampling periods, and models.

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A Relationship to previous measurement approaches

Table 6: Selected antecedents and the measurement distinction. This is a focused comparison, not a systematic literature review or an absence claim about every prior study.

Study	Measurement	Relation to this study
Bhandari et al. [2]	Prompted distances for 93 contiguous-US cities; five-beam decoding; planar MDS and geographic control.	Direct reconstruction precedent. Here: repeated stochastic requests, global spherical fitting and an absolute-error-preserving control.
Karimi and Janowicz [9]	Illustrative English–Persian estimates; Google Maps routing references.	Multilingual distance precedent; different reference target and experimental coverage.
Decoupes et al. [5]	Geographic coverage, country knowledge and semantic-distance distortion across ten models.	Embedding and knowledge indicators differ from repeated explicit geodesic estimates.
MultiGlobeQA [3]	Geospatial questions across 14 function families and 17 languages.	Broader multilingual task evaluation; our contribution concerns repeated numeric judgments and geometric consistency.

B Response dispersion and complete-pair comparisons

Table 7: Repeated-response variation. SD and coefficient of variation (CV) are averaged equally across all 4,950 pairs using accepted values; identical counts use each condition's 4,949 ten-valid-response pairs.

Instructions	Mean SD (km)	Mean CV (%)	Ten identical
English	116.4	1.50	139
Arabic	125.2	1.63	123
Chinese	121.5	1.57	64

C Protocol and experiment identifiers

The exact prompts are stored in `results/exact-language-prompts.json`. The prompt-family identifier is `great-circle-formatted-number-v2`; the parser identifier is `ascii_decimal_v2`. The English instructions are:

Table 8: Complete-pair comparisons. MAE change is second minus first condition; positive means larger pair-median error. Both p columns are Holm-adjusted across six tests of the strong response-distribution null.

Conditions	Disagreement (km)	MAE change (km)	p_T	p_A
English / Arabic	141.3	+18.1	0.0012	0.0012
English / Chinese	157.7	+6.3	0.0012	0.0012
Arabic / Chinese	165.2	-11.8	0.0012	0.0012

> Estimate the straight-line great-circle distance in kilometres between {city_a}, {country_a} and {city_b}, {country_b}. Return only your best numerical estimate in kilometres. Do not explain your reasoning. Use only digits 0–9, with a period if a decimal separator is needed. Do not include thousands separators or units.

The dash above is typographic; the Unicode supplement preserves the prompt string. All three runs use ten samples per pair, temperature 1, top-p 0.95, maximum output 128 tokens, and minimal thinking. Full parameter manifests, timing records, and source hashes accompany the exported data.

Table 9: Experimental conditions and export identifiers.

Instructions	Experiment ID	Analysis ID
English	`3b382690950b9511f5806ebc`	`114bbc2e2e5a9331`
Arabic	`b2e583773ba8111b0607ae88`	`3ef8ee3db9b488b1`
Chinese	`2ee0bcb921b94263ac26e635`	`bae2d55a858255c4`

The normalized-record dataset digest is

`087c6e422985dddcaaf72b9c1f85ce8d08e43bb2e5fb26eff19f0759ad2dd4c4`

The input lock separately records the CSV byte hash. These identifiers represent different serialized objects and need not match.

D Operational groups and distance balance

The Arab States regional definition uses the dated 22-country list in the 2021 ITU/UNU report [8], intersected with the 100-capital dataset: Abu Dhabi, Manama, Djibouti, Algiers, Cairo, Baghdad, Amman, Kuwait City, Beirut, Rabat, Nouakchott, Muscat, Doha, Riyadh, and Tunis. The Chinese case set is explicitly Beijing and Singapore, supported by the institutional and census sources [21, 20]. It is not a claim that Mandarin is absent elsewhere or predominant in every included setting. The source supplement records included and absent IDs, URLs, dates, and limitations.

Table 10: Regional composition and descriptive distance standardization. Distances are median WGS84 distances in km. Standardized gain and interaction apply identical pooled quintile weights to associated and other pairs; all five bins have observations in both groups.

Group	One endpoint	Both	Associated distance	Other distance	Std. gain	Std. interaction
Arabic	1275	105	5451	8474	-8.6	+13.6
Chinese	196	1	8659	7562	-12.0	-6.0

The complete-case rule excludes neither group's associated pairs. Thus the omitted observations affect only the other-pair component of these regional sensitivities. Group choice and coarse distance adjustment are exploratory; they do not control all geographic or linguistic confounding.

E Empirical-bootstrap audit and calibration

The retained audit uses 9,999 within-pair, within-condition empirical resamples and basic intervals [4, 18]. Its probability columns and intervals are not used for the paper's regional conclusions. For n observed errors with unbiased variance s^2, one draw from the empirical distribution has variance $(n-1)s^2/n$. The mean of n such draws has conditional variance $(n-1)s^2/n^2$. Scaling centered bootstrap mean deviations by $\sqrt{n/(n-1)}$ matches the usual variance estimate s^2/n. For a linear contrast $\hat{\theta} = \sum_c a_c \bar{e}_c$ over independent pair–condition cells with common $n = 10$, the corresponding variance is $\sum_c a_c^2 s_c^2/n$. This is a second-moment identity, not a proof of coverage.

For corrected deviations δ_b, basic intervals are $[\hat{\theta} - q_{.975}(\delta), \hat{\theta} - q_{.025}(\delta)]$. Approximate two-sided probabilities count $|\delta_b| \geq |\hat{\theta}|$, using the plus-one convention. The regional audit saves exceedance counts: zero for both Arabic statistics, 23 for the Chinese associated gain, and 2,146 for its interaction among 9,999 draws. Holm-adjusted values are 0.0004, 0.0004, 0.0048, and 0.2147. The first two are at the current Monte Carlo resolution. Corrections are specific to this four-test family and do not account for the preceding exploratory selection process.

We tested the same implementation on four explicitly synthetic, equal-mean scenarios, separate from all model observations: regular tied values, unequal variances, an unseen rare positive tail, and zero variance. Each used 240 independent outer datasets, 40 fixed pairs, ten observations per condition, and 499 inner bootstrap draws. Coverage ranged from 94.2–96.3% in the regular tied case and 93.8–95.0% with unequal variances; corresponding four-test familywise rejection rates were 4.6% and 3.3%. The deterministic case had full coverage and no rejection. In the deliberately rare-tail case, associated-gain coverage collapsed to 3.3–7.5% and familywise rejection reached 100%. These simulations diagnose possible failure; they neither identify the real response distribution nor validate universal nominal coverage. Monte Carlo standard errors, Wilson intervals, complete scenario definitions, and seeds accompany the report.

Conditional empirical intervals for associated gains are $[-11.7, -6.1]$ km for Arabic and $[-17.6, -3.8]$ km for Chinese. The interaction intervals are $[9.7, 16.7]$ and $[-11.6, 2.6]$ km. They are presented alongside the tail limitation, not as evidence that unseen rare responses are absent. The analytic-versus-bootstrap standard-error agreement remains an implementation check only.

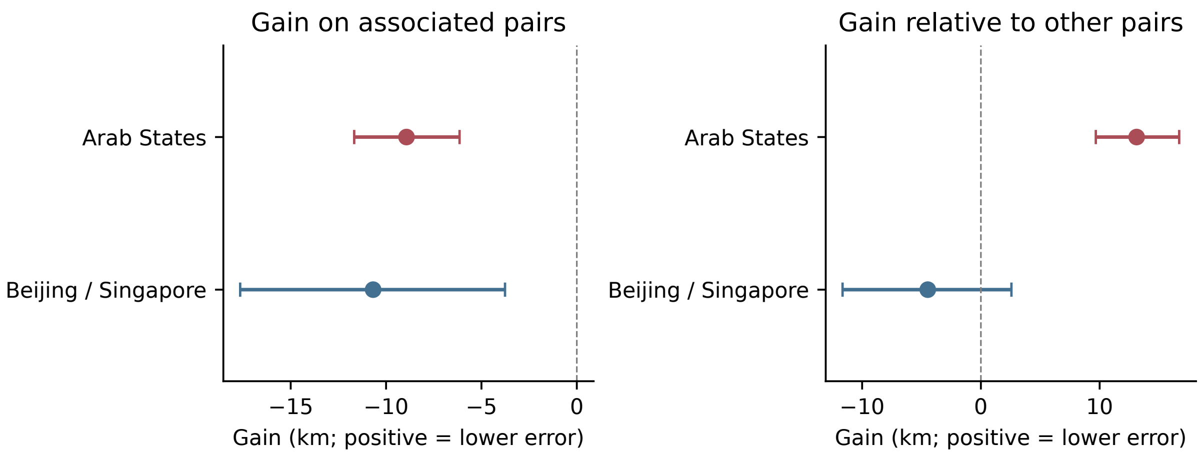

Figure 4: Retained methodological audit: regional point estimates and pointwise empirical-bootstrap intervals. Positive gain favors the translated prompt; positive interaction alone need not do so. These intervals are not used for regional conclusions. The rare-tail scenario in this appendix demonstrates severe undercoverage; the displayed intervals do not establish coverage of unseen response modes or variation across capitals and prompts.

F Exclusion and algorithm sensitivities

Table 11: Pair-median MAE (km) under recorded quality-control alternatives. Range-retained means include readable over-range numerical answers while continuing to exclude malformed text. No observations are invented or repaired.

Instructions	All pairs, accepted	Complete pairs	All pairs, range retained
English	215.86	215.77	216.12
Arabic	233.98	233.84	233.99
Chinese	222.02	222.05	222.02

Complete accepted pairs number 4,948; retaining numerical range violations yields 4,949 complete pairs, still excluding Brussels–Paris. All raw observations remain fixed. These descriptive alternatives answer a different selection question and are not additional preregistered tests.

F.1 Embedding residuals

Table 12: Distance-normalized stress for classical, metric planar, and spherical reconstructions; non-metric disparity Stress-1 is deliberately separated. Planar kilometer residuals use coordinates before display alignment. Lower values across the first three columns describe their stated in-sample distance objective, not model selection or equivalent dimensional constraints.

Instructions	Classical 2D	Metric 2D	Sphere	Non-metric disparities
English	0.2214	0.1569	0.0299	0.1415
Arabic	0.2246	0.1583	0.0350	0.1412
Chinese	0.2258	0.1538	0.0321	0.1391

The supplement retains all cosine eigenvalues and per-start diagnostics, including the rank-three positive-semidefinite checks on actual spherical arcs. Those controls separate implementation validity from whether judgment distances satisfy the spherical model.

G Complete 100-capital sample

ISO	Capital	Country
AE	Abu Dhabi	United Arab Emirates
AO	Luanda	Angola
AR	Buenos Aires	Argentina
AT	Vienna	Austria
AU	Canberra	Australia
BD	Dhaka	Bangladesh
BE	Brussels	Belgium
BF	Ouagadougou	Burkina Faso
BH	Manama	Bahrain
BO	Sucre	Bolivia
BR	Brasília	Brazil
BW	Gaborone	Botswana
CA	Ottawa	Canada
CD	Kinshasa	Democratic Republic of the Congo
CH	Bern	Switzerland
CI	Yamoussoukro	Ivory Coast
CL	Santiago	Chile
CM	Yaoundé	Cameroon
CN	Beijing	China
CO	Bogotá	Colombia
CR	San José	Costa Rica
CU	Havana	Cuba
CZ	Prague	Czechia
DE	Berlin	Germany
DJ	Djibouti	Djibouti
DK	Copenhagen	Denmark
DZ	Algiers	Algeria
EC	Quito	Ecuador
EG	Cairo	Egypt
ES	Madrid	Spain
ET	Addis Ababa	Ethiopia
FI	Helsinki	Finland
FJ	Suva	Fiji
FR	Paris	France
GB	London	United Kingdom
GH	Accra	Ghana
GR	Athens	Greece
GT	Guatemala City	Guatemala
HU	Budapest	Hungary
IE	Dublin	Ireland

ISO	Capital	Country
IN	New Delhi	India
IQ	Baghdad	Iraq
IR	Tehran	Iran
IS	Reykjavík	Iceland
IT	Rome	Italy
JO	Amman	Jordan
JP	Tokyo	Japan
KE	Nairobi	Kenya
KH	Phnom Penh	Cambodia
KR	Seoul	South Korea
KW	Kuwait City	Kuwait
LA	Vientiane	Laos
LB	Beirut	Lebanon
MA	Rabat	Morocco
MG	Antananarivo	Madagascar
MN	Ulaanbaatar	Mongolia
MR	Nouakchott	Mauritania
MX	Mexico City	Mexico
MY	Kuala Lumpur	Malaysia
MZ	Maputo	Mozambique
NA	Windhoek	Namibia
NG	Abuja	Nigeria
NL	Amsterdam	Netherlands
NO	Oslo	Norway
NP	Kathmandu	Nepal
NZ	Wellington	New Zealand
OM	Muscat	Oman
PA	Panama City	Panama
PE	Lima	Peru
PG	Port Moresby	Papua New Guinea
PH	Manila	Philippines
PK	Islamabad	Pakistan
PL	Warsaw	Poland
PT	Lisbon	Portugal
QA	Doha	Qatar
RO	Bucharest	Romania
RS	Belgrade	Republic of Serbia
RU	Moscow	Russia
RW	Kigali	Rwanda
SA	Riyadh	Saudi Arabia
SB	Honiara	Solomon Islands
SE	Stockholm	Sweden
SG	Singapore	Singapore

ISO	Capital	Country
SN	Dakar	Senegal
TH	Bangkok	Thailand
TN	Tunis	Tunisia
TO	Nuku’alofa	Tonga
TR	Ankara	Turkey
TZ	Dodoma	United Republic of Tanzania
UA	Kyiv	Ukraine
UG	Kampala	Uganda
US	Washington, D.C.	United States of America
UY	Montevideo	Uruguay
UZ	Tashkent	Uzbekistan
VN	Hanoi	Vietnam
VU	Port Vila	Vanuatu
WS	Apia	Samoa
ZA	Pretoria	South Africa
ZM	Lusaka	Zambia
ZW	Harare	Zimbabwe